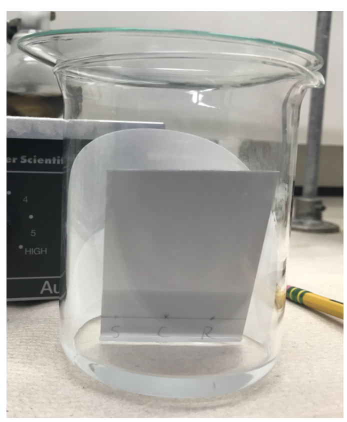
Chem 0345/0330 – Organic Lab
Activity 4: TLC
Extra Practice Problems

1. Classify each material below as a stationary phase or mobile phase. You may not have explicitly discussed each of the materials below; use your knowledge of how stationary and mobile phases are used in chromatography to answer this question.

Material	**Stationary or Mobile Phase**
Silica	
Hexane	
Ethyl acetate	
C18-modified silica	
Dichloromethane	
Alumina (Al_2O_3)	
Methanol	
Diethyl ether	
Glass	

2. Label the TLC setup below with each of the following components and describe each one's purpose:



   A. Lid

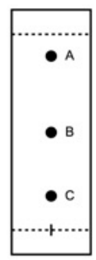
   B. Filter paper

   C. Stationary phase

   D. Mobile phase

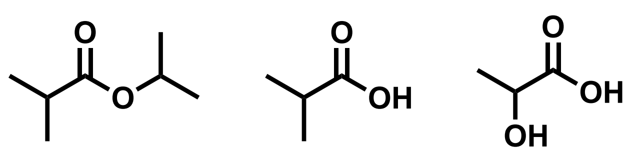
3. Briefly describe the process for running a TLC.

4. What is the R_f for each spot below? What can you say about the relative polarity and retention time of each spot?

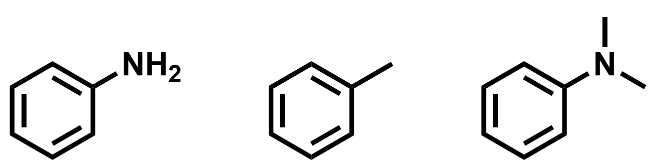


   A. Spot A
   B. Spot B
   C. Spot C

5. Assign the likely identity of each spot in Question 4 to the molecules below:



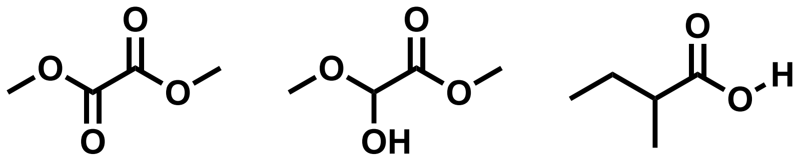
6. Which of the following statements is true about the three molecules below? Select all that apply.



A B C

   A. Molecule A will have a smaller retention time than Molecule B
   B. Molecule A will have a larger R_f than Molecule B
   C. Molecule B will have a smaller retention time than Molecule C
   D. Molecule B will have a smaller R_f than Molecule C
   E. Molecule A will have a smaller Kp than Molecule B

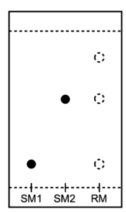
7. Which of the following statements is true about the three molecules below? Select all that apply.



A B C

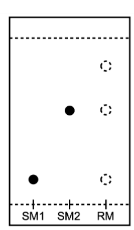
A. Molecule A will have a smaller retention time than Molecule B
B. Molecule A will have a larger R_f than Molecule B
C. Molecule B will have a smaller retention time than Molecule C
D. Molecule A will have a smaller R_f than Molecule C
E. Molecule A will have a smaller Kp than Molecule B

8. A reaction using starting material 1 (SM1) and starting material (SM2). After 1 hour, a TLC of the starting materials and the reaction mixture (RM) was run. Which one of the following statements is true? You can assume that equal amounts of starting materials were used.



A. The reaction is complete because all starting materials have been consumed.
B. The reaction is not complete and should be run for additional time.
C. The reaction is complete even though all starting materials have not been consumed.
D. The reaction is not complete because the starting materials are impure.

9. A reaction using starting material 1 (SM1) and starting material (SM2). After 1 hour, a TLC of the starting materials and the reaction mixture (RM) was run. Which one of the following statements is true? You can assume that equal amounts of starting materials were used.



A. The product is more polar than both SM1 and SM2.
B. The product is more polar than SM1 but not SM2.
C. The product is less polar than both SM1 and SM2.
D. The product is less polar than SM1 but not SM2.